Evaluating Ratings from Large Language Models: A Simulation-Based Workflow

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Abstract

Large Language Models (LLMs) are increasingly used as research tools to facilitate fast and automated ratings of text features. In psychological studies, they have been used to quantify the degree to which stimulus materials reflect certain psychological constructs. However, the application of LLMs entails a high degree of flexibility regarding prompt design (e.g., instruction details and examples) and model specification (e.g., model family, size, and configuration), which can produce divergent results and threaten the robustness and generalizability of conclusions. To navigate the multiverse of possible choices, we develop a systematic workflow for evaluating the quality of LLM-generated ratings across diverse model and prompt specifications. Motivated by generalizability theory, the workflow distinguishes between construct variance across stimulus items, method variance due to model and prompt choices, and residual variance across repeated iterations. To guide researchers through the planning, execution, and reporting of LLM simulation studies, we adapt the ADEMP (Aims, Data-generating mechanism, Estimands and targets, Methods, Performance measures) framework, originally developed for methodological simulation research. We develop a template supporting two complementary validation strategies: assessing consistency across LLM specifications based on generalizability theory via estimated variance components, and external validation against a benchmark (e.g., human ratings). We illustrate the workflow in a preregistered case study using local, open-weight LLMs which examines the influence of model choice, response format, and prompt examples on the quality of valence and arousal ratings for multi-word expressions.

Nontechnical Abstract.

Researchers in psychology increasingly use Large Language Models (LLMs) to automatically rate stimulus materials—for example, to determine how positive or emotionally activating a word or phrase is. While this approach promises fast and cost-effective alternatives to collecting human ratings, LLMs offer enormous flexibility: researchers must choose among many available models and decide how to phrase prompt instructions. It is usually unclear to which degree these choices influence the results

obtained with LLMs, yet most studies rely on a single model with a single prompt. This article provides a simulation-based workflow and a corresponding template to systematically test how robust LLM-based results are across the multiverse of different models and prompt designs, and to assess the generalizability of LLM-generated ratings. A case study evaluating valence and arousal ratings demonstrates that even smaller, freely available models can produce ratings comparable to those of leading commercial models. The proposed workflow enables applied researchers to make informed decisions about LLM use, improve the transparency of their methods, and increase confidence that LLM-generated ratings are sufficiently consistent to support substantive conclusions.

Keywords: Stimulus ratings, G-theory, LLMs, multiverse analysis, simulation studies.

Evaluating Ratings from Large Language Models: A Simulation-Based Workflow

Large language models (LLMs) are increasingly used in psychological research to analyze and generate verbal data and stimulus materials. Examples include analyzing large quantities of qualitative data (e.g., Chew et al., 2023; Player et al., 2025; Rathje et al., 2024; Smirnov, 2025), automating item scale construction (e.g., Barends & de Vries, 2025; Fyffe et al., 2023; Hommel et al., 2022), generating ratings of words, expressions, and phrases regarding psychological constructs (e.g., Hommel, 2023; Martínez et al., 2024), or using LLMs to simulate and estimate human behavior and judgments (e.g., Loru et al., 2025; von der Heyde et al., 2025). In light of the numerous potential applications, it has been argued that LLMs have the capacity to facilitate various steps in the research process and to advance cognitive science (Feuerriegel et al., 2025; Wulff & Mata, 2025a).

The present paper focuses on the application of LLMs to analyze and rate text stimuli. For example, Martínez et al. (2024) used GPT-4o to generate ratings of valence (i.e., the pleasantness of a stimulus) and arousal (i.e., the intensity of emotional activation a stimulus elicits; Warriner et al., 2013) for multi-word expressions such as "hit rock bottom," "fall in love," or "ice cream cone." A defining feature of such tasks is that they involve a corpus of text stimuli (e.g., words, questionnaire items, sentences, social media posts, vignettes, etc.). The stimuli are assumed to differ with respect to relevant features and LLMs are used to rate the stimuli automatically with respect to psychological constructs (arousal, valence), linguistic properties (grammatical complexity, concreteness), or other quantitative or qualitative information (topic modeling, presence or absence of certain informational cues). Researchers using LLMs for such tasks are necessarily faced with the question of how to evaluate the quality of the generated ratings.

Challenges of Using LLMs in Research

LLMs offer many impressive capabilities but also come with notable limitations for research use. Both small and large models are known to be susceptible to hallucinations (Huang et al., 2025; Reddy et al., 2024) and they may generate synthetic response behavior that does not resemble that of human participants (Argyle et al., 2025;

Hullman et al., 2026). Furthermore, LLMs may fall short in performing even seemingly simple tasks such as word counting (Markowitz, 2024). Despite their capacity to emulate plausible human responses in specific tasks, LLMs frequently do not succeed in replicating the full spectrum of human behavior. This limitation is exemplified by the substantial reduction in variability of responses observed in LLM outputs compared to humans (Abdurahman et al., 2024; Martínez et al., 2024). Such issues and shortcomings have the potential to result in errors when relying on LLM output for making statistical inferences and drawing substantive conclusions (Hullman et al., 2026). Accordingly, it is imperative not to interpret LLM output at face value, as it has the potential to be misleading or false. Utilizing LLMs in research responsibly necessitates the implementation of rigorous quality control measures (e.g., Argyle et al., 2025; Knöchel et al., 2025; Nestler et al., 2026).

When aiming to use LLMs for a particular scientific task, researchers are faced with a multitude of different models and approaches. In light of the abundance of available LLMs, the selection of a particular setup often involves arbitrary choices, which introduces researcher degrees of freedom. Models differ in important aspects such as number of parameters, architecture, hyperparameter settings, and how and with which text corpora they have been trained (Argyle et al., 2025; Hussain et al., 2024). Emerging applications of LLMs often rely on a single, promising LLM (e.g., Barends and de Vries, 2025; von der Heyde et al., 2025). The selection of a particular model might be based on preliminary pilot testing or practical considerations such as accessibility and ease of use. Output quality can be increased by systematic, iterative testing and may benefit from advanced methods such as fine-tuning (for a tutorial, see Conde et al., 2026). Reliance on a single model does not necessarily constitute a problem if its performance has been validated for the task at hand. For example, Martínez et al. (2024) used GPT-4o to obtain ratings of valence, arousal, and concreteness for thousands of single words and multi-word expressions. To validate their approach, they showed that the LLM ratings correlated highly both with existing human norms and with newly collected human ratings that were not part of the LLM training data. However, once LLM output is used to draw scientific conclusions, it is important to investigate whether the results are robust

across different models. This is especially relevant given that new model variants with potential benefits (e.g., higher accuracy, increased efficiency, or reduced energy consumption) are continuously being developed.

A related issue is that LLM outcomes usually vary even when providing the same prompt repeatedly due to stochasticity inherent in model architecture (Demszky et al., 2023; Knöchel et al., 2025). However, computational demands might be too high to repeat the same prompt inputs multiple times to study variability and to attain stable estimates. As a remedy, one may enforce deterministic output by setting the temperature hyperparameter to zero. However, such an approach artificially ignores the uncertainty of the underlying, stochastic model as it merely picks the most likely response option irrespective of whether it is clearly or only slightly preferred to the second-best option (Abdurahman et al., 2024; Martínez et al., 2024). Even if temperature is set to zero, this does still not guarantee deterministic output for some LLMs (Abdurahman et al., 2025). Research investigating this problem often identified settings of other parameters, such as the batch size, as the cause for issues with reproducibility (Coqueret et al., 2026; Yuan et al., 2025). Furthermore, minor variations in prompt design may result in divergent outputs, thereby casting doubt on the validity, robustness, and generalizability of LLM output (e.g., Baumann et al., 2025; Cummins, 2026; Feuerriegel et al., 2026; Sclar et al., 2024).

Researchers have also criticized the usage of commercial LLMs since it is contingent upon proprietary services that may not be available to all researchers equally and do not adhere to open science principles (Balluff et al., 2026; Conde et al., 2026; Guest & van Rooij, 2025; Hussain et al., 2024). As a remedy, researchers can use locally running, open LLMs for which the underlying model architecture and the parameter weights are known and available (Hussain et al., 2024), e.g., by using Ollama (Ollama, 2026). Further potential pitfalls of using LLMs have been reviewed by Argyle et al. (2025) and Wulff and Mata (2025a).

A Simulation-Based Workflow for Evaluating LLM Ratings

The overview of limitations above shows that results obtained with one particular (commercial) LLM may not be robust to modifications of model and prompt parameters. In line with good scientific practice, we recommend that researchers use open-weight LLMs and report the employed procedures and results transparently to ensure reproducibility and replicability of the research process (Abdurahman et al., 2025; Feuerriegel et al., 2025; Hussain et al., 2024; Knöchel et al., 2025). More importantly, it is essential to study the robustness and generalizability of LLM output across the multiverse of possible decisions regarding model choice, configuration, and prompt design (Hussain et al., 2024; Steegen et al., 2016). Our evaluation framework thereby complements workflows focusing on method development via iterative improvement of LLM ratings (Conde et al., 2026).

Generalizability theory (Cronbach, 1972) provides a natural framework for analyzing ratings of different raters (e.g., LLM specifications) for a set of items (e.g., text stimuli) by partitioning the observed variance into systematic and error components. By distinguishing different sources of measurement error (termed *facets*), generalizability theory enables researchers to quantify the contributions of different factors. Besides reducing unexplained variability, this provides a better understanding whether measurements generalize across different raters, items, and/or contexts (Brennan, 2001). Generalizability theory and variance-decomposition approaches have recently been applied for specific LLM tasks, such as automated essay scoring (Song et al., 2026) and user intent extraction (Kunievsky & Evans, 2026), as well as for AI benchmarks (Hardy et al., 2026) , and have been advocated more generally for establishing the reliability of AI-based scores (Speer et al., 2026).

Generalizability theory is particularly well-suited for studying the quality of LLM ratings in simulation studies, where output variability arises from multiple sources (e.g., prompt structure, model architecture, or stochasticity of the generation process). While empirical studies with human participants are often constrained by a relatively small number of measurements, research on LLM ratings enables the implementation of a fully

crossed design with many facets to study the factors driving output variability. In a simulation, different models can be asked to rate all items in each combination of various conditions. Across repeated prompts, LLMs can be called in a fresh context—free of prior conversation history—to ensure stochastic independence between measurements and conditions. This high degree of methodological control enables a principled psychometric evaluation of LLM rating quality using generalizability theory.

From a pragmatic perspective, we recommend that researchers in psychology routinely employ a simulation-based workflow to evaluate the robustness of LLM-generated ratings for stimulus materials (e.g., words, sentences, test items, or vignettes), especially when there is no or little prior information regarding robustness of performance under different model and prompt parameters. When systematically varying model and prompt specifications in a factorial design, researchers can apply principles established for simulation studies in methodological research. Specifically, we adopt the ADEMP framework, which guides researchers in specifying the Aims, Data-generating mechanism, Estimands and targets, Methods, and Performance measures of a simulation study (Morris et al., 2019). We modified the ADEMP-PreReg template (Siepe et al., 2024), which provides a systematic structure for planning, analyzing, and reporting simulation studies that evaluate the performance of different statistical methods (Siepe et al., 2024). The template can also be used for preregistration, although the benefits and limitations of preregistering simulation studies remain debated (Siepe et al., 2026).

In the following, we develop a systematic simulation workflow to enable robust and transparent evaluation of LLM-generated ratings. We adopt generalizability theory and propose concrete procedural steps for evaluating the quality of LLM ratings of verbal materials. The modified ADEMP template for LLM simulation studies focuses on establishing a universally applicable, systematic workflow that can be utilized across diverse domains of psychological and social science research. We will refer to this type of evaluation study as *LLM simulation*. Note that other studies have employed similar terms (e.g., “LLM social simulations”; Anthis et al., 2025) to refer to research in which LLMs are employed to emulate hypothetical responses of human participants (so called

"silicon samples", Cummins, 2026). In contrast, our use of the term LLM simulation refers to the systematic evaluation and variation of methodological design choices when eliciting LLM ratings.

A complementary line of work has proposed prompting guidelines for various research purposes (e.g., Conde et al., 2026; Fang et al., 2026; Giray, 2023; Ziems et al., 2024). For example, prompts should be clear, precise and contain definitions of the target constructs. Furthermore, prompt instructions may be accompanied by example cases illustrating the task, which is referred to as few-shot prompting, and prompts may assign roles to the LLM (e.g., "You are a statistics expert," Conde et al., 2026; Fang et al., 2026). However, none of these prompting approaches is a safeguard against perturbations in the prompts, as studies have repeatedly found that subtle changes in prompts can elicit major variations in LLM responses (Alahmari et al., 2025; Sclar et al., 2024). Moreover, specific recommendations are necessarily limited in scope, since they may not transfer across research domains and tasks, across models, or across time, given the rapid development of new models. In contrast, our proposed framework and template are sufficiently general to evaluate a wide range of LLM pipelines for diverse research questions. The two approaches complement each other. While prompting guidelines suggest promising prompt designs, our evaluation workflow allows researchers to test whether these designs actually improve the quality of LLM ratings in a given application by including them as method factors.

Comparing Statistical and LLM Simulation Studies

In methodological simulation studies, researchers typically define a data-generating mechanism to simulate a large number of synthetic datasets. Next, different statistical methods are applied to each dataset. The employed methods are then evaluated using different performance measures that reflect how well each method achieves its goal (Morris et al., 2019). Using this approach, researchers aim to draw conclusions regarding properties of different methods and make recommendations for applied data analysis under a specified set of conditions (Siepe et al., 2024).

It is useful to adopt this general approach to study the multiverse of possibilities

in LLM prompting. An LLM simulation study is characterized by systematically varying procedural details when generating ratings using different model specifications and prompt structures. The quality of the resulting item ratings is then evaluated using one or more performance measures. This allows researchers to quantify which design choices are critical for obtaining LLM-generated ratings in a specific application scenario. For example, researchers can quantify the consistency between various LLMs' ratings or assess whether the inter-rater reliability with human ratings depends on specific aspects of the prompting procedure.

Despite many similarities, simulation studies evaluating statistical methods also differ from those evaluating LLM-generated ratings in important ways. In LLM simulations, it is particularly important to distinguish three types of variability that affect the obtained output. First, the variance of LLM ratings depends on the latent variability in the rated feature dimension across items. This type of variance constitutes *construct variance*, which is the target of generated ratings. To ensure a sufficient amount of construct variance, the items to be rated should be a representative sample from the target domain of interest. The definition of the target domain determines the generalizability of the observed findings to other items from this domain that have not been studied (Brennan, 2001). Second, systematically varying prompts and models enables researchers to estimate *method variance* in observed LLM ratings. [1] Third, LLM ratings are inherently stochastic and can vary even when using the same model and prompt for the same item repeatedly. This type of variability across iterations is particularly important for judging a model's degree of *residual variance* (i.e., stochastic error), which determines the stability of LLM responses to a particular task. Note that low variability of LLM output to repeated, identical prompting does not imply that the provided response is correct. Moreover, the variability of different models is only comparable if parameters that influence stochasticity (e.g., temperature, top-p) are set to

[1] We adopt the terms construct variance and method variance as accessible labels for the more formal terms universe score variance and facet-related error variance components recognized in generalizability theory (Brennan, 2001; Cronbach, 1972). The term method variance is borrowed from the multitrait–multimethod tradition (Campbell & Fiske, 1959). The term residual variance is used as in generalizability theory and has to be distinguished from *error variance* (see the Performance Measures section of the case study).

the same values, although even identical settings may affect models differently.

Another important distinction between methodological simulation studies and LLM simulations concerns the data-generating mechanism. When employing LLMs, the data-generation mechanism refers to the process of mapping the provided input (i.e., a specific prompt version providing instructions and the target item to be rated) to the output (e.g., ratings about psychological variables) by running a certain model configuration. Many details of this process may be unknown or not clearly specified (e.g., the model architecture or parameter values). Many state-of-the-art models have billions of parameters (e.g., between four and 70 billion parameters for typical open source models), which interact in complex ways, making it particularly difficult to understand the inner processing of LLMs (Feuerriegel et al., 2025). In contrast to statistical simulation studies, LLM simulations do not aim at recovering some known, "true" parameters of the (underspecified) data-generating model. Instead, they aim at evaluating the quality of the obtained LLM output with respect to the task of interest (e.g., whether LLM ratings provide a sufficiently close approximation to a gold standard).

ADEMP Template for LLM Simulation Studies

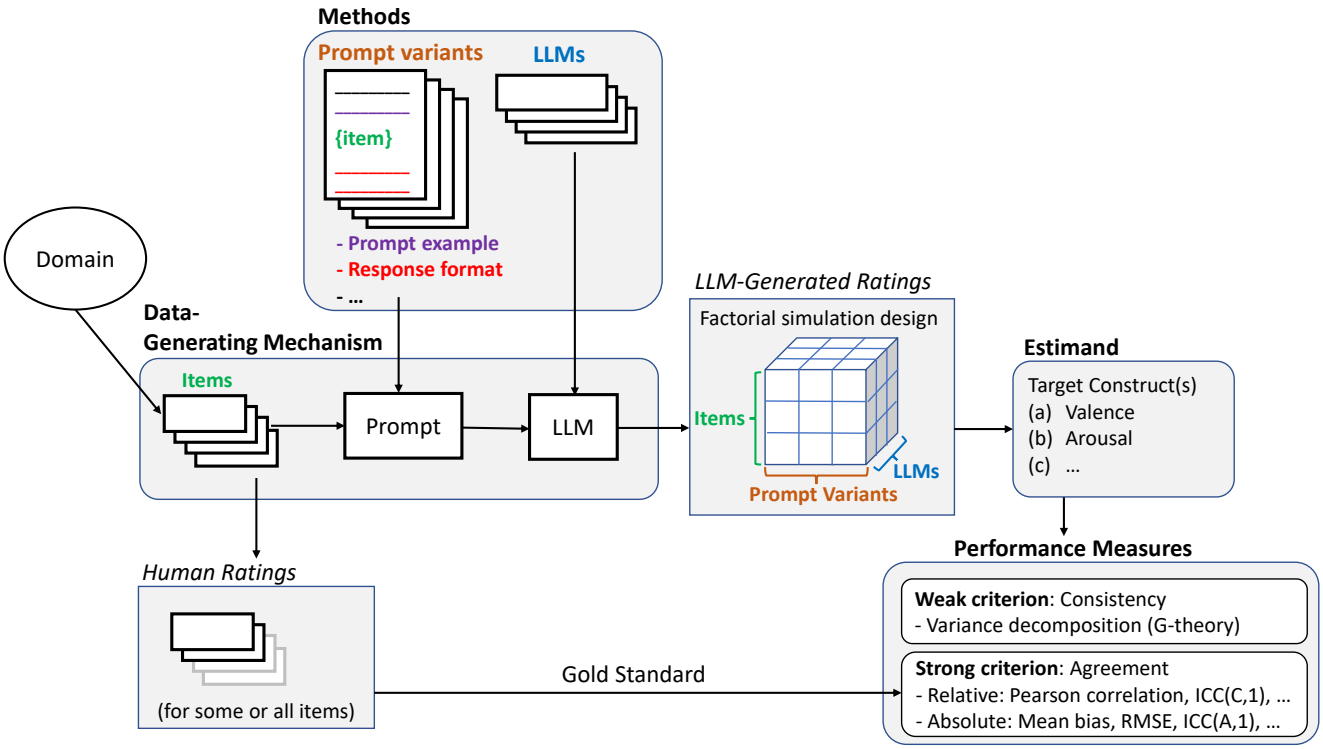
The modified ADEMP framework for LLM simulation studies, illustrated in Figure 1, addresses the specific requirements of evaluating LLM ratings of psychological constructs for text stimuli based on generalizability theory. The corresponding ADEMP template, available at https://github.com/TImA97/ADEMP-PreReg-for-LLMs (or https://doi.org/10.5281/zenodo.23762870 on Zenodo), enables researchers to plan and potentially preregister the evaluation of LLM rating quality in pilot or validation studies, either with or without human ratings (or any other criterion variable). The template also guides the transparent reporting of results.

Aims

The Aims section describes the overarching goal of the LLM simulation study. Researchers are typically interested in evaluating the ability of different LLMs to fulfill a particular rating task. Sometimes, human ratings of the materials are available as a benchmark for validation (e.g., by computing a correlation between human and LLM

Figure 1

ADEMP framework for LLM simulation studies.



Note. Adapted from the ADEMP (Aims, Data-generating mechanism, Estimands and targets, Methods, Performance measures) framework for statistical simulation studies (Morris et al., 2019).

ratings). In initial research stages, however, human ratings may not be available, which calls for different strategies for the preliminary validation of the LLM output. Depending on whether an external validation criterion is available, one can distinguish two aims. The illustrated ADEMP workflow in Figure 1 highlights the distinction between assessing LLM ratings in terms of consistency and in terms of agreement with a gold standard.

Irrespective of whether human ratings are available as a benchmark, LLM simulations may aim at evaluating the robustness and consistency of ratings across various LLM specifications. In psychometric terms, this reflects the amount of method variance in observed ratings due to asking multiple raters (i.e., different LLMs and prompt instructions). Potential aims might include conducting a variance decomposition of LLM ratings regarding items, models, and prompting approaches, or assessing inter-rater reliability between different LLMs or between the stochastic, repeated output of the same LLM. Variance decomposition offers the advantage of reducing the number of

comparisons and eliminating the need to define one of the LLM variants as a benchmark for comparison. The variance explained by method factors such as the employed model or prompt structure should account for only a relatively small proportion of the variance compared to the construct variance across items. Such a variance decomposition allows researchers to assess whether variability—or, conversely, consistency—in LLM ratings is due to factors exogenous or endogenous to the model (Sundh et al., 2025).

If human ratings from experts or participants are already available or are planned to be collected for some or all of the items, LLM simulations may aim at investigating the relative or absolute agreement of LLM ratings with benchmark ratings (e.g., Fang et al., 2026; Martínez et al., 2024). If LLMs are used to extract information about an objective criterion (e.g., the factual truth of statements), the LLM simulation aims at assessing criterion validity, which can be estimated using measures of absolute agreement. The aim of assessing how well LLMs map to a gold standard mirrors the goal of traditional simulation studies, in which a statistical method is evaluated with respect to its performance in recovering known, true parameters (Morris et al., 2019).

Data-Generating Mechanism

When using LLMs, the data-generating mechanism is the process of mapping the prompt input to observable output as generated by a certain model configuration. However, the underlying process of how exactly the input is transformed, processed, and analyzed by an LLM is often unknown and difficult to grasp (Abdurahman et al., 2024), even if it can be formally specified with mathematical equations and parameter values. What can, however, be specified when generating LLM ratings are analytical decisions at a more abstract level: (a) the selection or sampling of stimulus materials from the target domain and (b) the procedures employed to obtain LLM ratings regarding psychological features of interest. Whereas the former aspect is relevant for the generalizability to other, new items from the target domain, the latter matters for the robustness and generalizability across different LLM specifications. Accordingly, the Data-Generating Mechanism section contains information both about the type and selection of stimulus materials rated by the LLMs, and about the exact procedures of how LLM ratings will be

elicited.

The stimulus materials used for LLM prompting (i.e., the corpus) can include statements, questions, vignettes, or items from a personality inventory. First, researchers must define the target domain of the psychological (or linguistic) construct they wish to investigate. Second, they need to decide on how to sample or construct stimuli that are representative for this domain (Cronbach, 1972). The definition of the corpus and the selection of items to be included directly shape the scope and generalizability of findings about LLM-extracted features. If the corpus consists only of statements assessing a particular psychological construct, the results of the LLM performance evaluation are informative only for comparable materials for the same construct. Ideally, one would draw a true random sample of items from a well-defined target domain (Brennan, 2001). While this is practically difficult or even impossible, researchers should ensure that the verbal materials cover a sufficiently broad and heterogeneous subspace of the target domain to increase the generalizability of conclusions. What constitutes sufficiently diverse data as input depends on the research question.

Initial attempts at extracting features via LLM prompting will likely involve user-friendly, interactive chat interfaces. However, to render the generation of LLM ratings systematic, efficient, and reproducible, it should proceed in an automated and standardized manner. Accordingly, the Data-Generating Mechanism section should describe all procedural details that are identical across all simulation conditions. This includes the general design and structure of the LLM prompt input, while specific components of the prompt that vary across conditions (e.g., whether and which examples are given) should be reported in the Methods section.

One should also specify relevant details of the software employed to obtain LLM ratings, for instance, whether a remote, commercial service, a locally hosted, open-weights model, or an interactive chat interface will be used. To ensure independence between ratings, it is important to use a fresh context when eliciting LLM responses (Conde et al., 2026). It should be reported how the models are accessed (e.g., via an API, that is, an application programming interface), and whether and how a specific response

format (e.g., a numeric Likert rating between 1 and 7) was ensured. For instance, it is possible to include specific instructions in the prompt to adhere to a certain response format or to use software features ensuring that the LLM output has a specific format, such as JSON (JavaScript Object Notation). Note that the choice for structured output against an open format itself constitutes a researcher degree of freedom and could be explicitly varied to assess to what extent LLM ratings depend on this methodological factor. We generally recommend using structured output for increasing standardization and software optimized for handling such output (e.g., the `type_object()` function family in the `ellmer` package in R; Wickham et al., 2025).

To ensure replicability of model calls, one should set random seeds, though this may not be possible for all models (Abdurahman et al., 2025). Reproducibility of LLM output also depends on hyperparameters such as temperature, top-k, and top-p that shape the distribution from which tokens are drawn. While top-k sampling uses only the k tokens with the highest probability, top-p sampling uses the smallest set of tokens with a cumulative probability of p (Coqueret et al., 2026). When prompting an LLM, even details such as the batch size can alter the obtained output. Full reproducibility is attainable only if a local model is run with fixed model and software versions, a fixed seed (or a temperature of 0), a controlled batch size, and identical hardware—conditions that are difficult to achieve in practice. Yet, it is important to minimize stochasticity where possible, because reproducibility in LLMs is not all-or-nothing. Fixing the seed, temperature and sampling parameters reduces unwanted variation and makes results more comparable. This is especially valuable when repeating prompts many times is computationally prohibitive (Coqueret et al., 2026; Yuan et al., 2025).

Estimands and Targets

The Estimands and Targets section describes the purpose of the ratings to be generated by clarifying the theoretical construct(s) being studied. In traditional, methodological simulation studies (Morris et al., 2019), the targets of interest are usually the unknown parameters of the data-generating mechanism, which are estimated with a statistical method. In contrast, the targets in LLM simulation studies are quantitative or

qualitative characteristics of the text stimuli in terms of the psychological variables of interest. Correspondingly, the estimands of LLM-generated ratings are the unknown (numeric or discrete) values of the items on the variables to be rated.

Methods

This section provides information about the different models and prompt variants that will be analyzed in the LLM simulation (cf. Figure 1). The chosen method factors should be accompanied with a clear description and justification to ensure that readers can judge the scope of the LLM simulation (i.e., which range of model specifications has been investigated). The combination of all method factors varied in the LLM simulation determines the simulation design and the number of simulation conditions.

It should be stated which LLMs will be used. LLMs differ in numerous aspects such as the architecture (often referred to as model family), the number of parameters (i.e., model size), the company or institution developing a model, the languages in which an LLM was trained, and whether information about the model weights and training procedure is publicly available. If possible, justify why the employed models were selected. The reasoning for choosing specific models may concern availability, expected costs, openness, and/or computational feasibility (e.g., maximum model size that can be run locally).[2]

A fully factorial design encompassing all combinations of LLM families and model sizes is often infeasible, as not all models are available across different size variants. Thus, we recommend including all investigated LLM variants as a single factor in the analysis as opposed to handling certain aspects (e.g., model family and model size) as separate factors. Furthermore, report the values of all model hyperparameters that may affect the output (e.g., temperature, top-p, top-k, context length, seed, batch size), including those that were not changed from their defaults. Default values may change between software versions, differ between models, and be overridden by the software used to access a model. For instance, in our case study, the documented default temperature of 0.8 in Ollama was

[2] For instance, a graphics card with 32GB VRAM can typically handle models up to about 30 to 35 billion parameters (maximum model size: 35B) depending on quantization.

replaced by a value of 1 because the models were accessed through an OpenAI-compatible endpoint (see Appendix B). This shows that it can be necessary to verify the effective settings rather than relying on the documentation. If hyperparameters are adjusted for each model, it is appropriate to treat all LLM variants as a single factor. Otherwise, if systematic variations of LLM settings (e.g., temperature, which controls the degree of stochasticity of the output) are of interest, these hyperparameters can be treated as separate method factors. For example, one may manipulate temperature and top-p in a factorial grid to study their relative impact on the variability and quality of LLM ratings.

The Methods section also describes the different prompt versions provided to the LLMs. Specify all components of the prompt input that vary across simulation conditions, such as different system prompts that may contain context about the purpose of the task, instructions for the LLM on which scale to return ratings, and whether examples are provided or not (zero-shot vs. few-shot prompting; Feuerriegel et al., 2025). Each of these prompt components defines a method factor. In a fully factorial design, distinct prompt versions are obtained by combining all levels of the different prompt components. If some of the combinations do not result in meaningful prompt instructions, one may use an incomplete factorial design or treat the subset of meaningful combinations as a single factor. To ensure that the comparison of LLM rating quality across models is valid, the same prompt versions should be used for all LLM versions.

Performance Measures

The Performance Measures section provides information on how the consistency and agreement of LLM ratings will be evaluated and how the uncertainty of this evaluation will be quantified. The section also concerns the handling of missing values and the interpretation of performance measures, as well as the number of iterations in each simulation condition. To avoid ambiguity, one may also specify equations for how these measures will be calculated.

The choice of specific performance measures depends on the aims of the investigation. If the aim is to analyze the consistency in terms of the proportion of variance in LLM ratings due to different LLM model versions or prompt instructions, it is

necessary to specify a suitable mixed-effects model for variance decomposition of the LLM simulation design. Generalizability theory provides a natural framework for such an analysis, which proceeds in two steps (Brennan, 2001). In the G study, a mixed model is used to estimate the variance components of the object of measurement (typically the stimuli), of all facets (e.g., model, prompt components), and of their interactions. In the D study, these components are combined into reliability coefficients for a specific measurement design, that is, for ratings averaged across a given number of conditions per facet. The generalizability coefficient $E\rho^2$ quantifies how consistently the relative ordering of the stimuli is reproduced across facet conditions, whereas the dependability coefficient Φ additionally takes differences in the absolute level of the ratings into account. Because a D study can be computed for any number of conditions per facet, researchers can also project how reliability will change if different numbers of models, prompt variants, or iterations are used. When planning the analysis, researchers should specify the mixed model, which facets are treated as random or fixed, how the universe score is defined, which coefficient is of primary interest, and which D study designs will be evaluated. The case study below illustrates these steps in detail.

Importantly, high consistency across simulation conditions does not imply that the output is correct or valid. High estimates of consistency can also emerge if LLMs systematically give incorrect or biased responses. Therefore, the weak validation criterion is not sufficient by itself to judge the quality of LLM ratings. Still, the approach is useful as a first step to identify methodological factors that shape LLM response distributions when an external criterion is not (yet) available.

If an external criterion is available for comparison (e.g., human ratings), the inter-rater reliability between human ratings and LLM ratings may be computed as the Pearson or rank-order correlation of ratings across all stimulus materials (Goodwin, 2001). The choice between measures of relative and absolute agreement should follow from the intended use of the LLM ratings. The distinction between relative and absolute agreement parallels the generalizability and dependability coefficients in generalizability theory, which concern the relative ordering and the absolute level of ratings, respectively

(Brennan, 2001).

Measures of relative agreement (e.g., Pearson or rank-order correlations, or the consistency intraclass correlation ICC(C,1); McGraw and Wong, 1996) are sufficient when LLM ratings are used to order stimuli, to select stimuli covering the range of a construct, or as covariates or predictors in statistical models, since these uses are unaffected by linear transformations of the ratings. Measures of absolute agreement (e.g., mean bias, root mean square error across items, or the agreement intraclass correlation ICC(A,1)) are required whenever the generated values themselves are interpreted, for instance, when LLM ratings are combined with existing human norms, when stimuli are selected based on cutoff values (e.g., to select "neutral" items), or when LLMs are used to predict or forecast observable quantities such as voting behavior (von der Heyde et al., 2025). Because high correlations do not preclude systematic shifts or a compressed range of LLM ratings (Abdurahman et al., 2024), researchers may additionally compare the distributions of LLM and human ratings (e.g., means and standard deviations).

For any measure, one should specify whether LLM ratings are averaged across prompt iterations before computing the mean across items. Furthermore, researchers may employ a rater non-inferiority criterion, demanding that the agreement between LLM and human ratings should be as high as the agreement between different human raters (Cummins et al., 2026). If a study targets specific comparisons between LLM configurations (e.g., whether a smaller local model agrees with human ratings as well as a commercial model), these comparisons can be tested statistically. Because both correlations share the same benchmark ratings, they are dependent, and tests for independent correlations are not appropriate. Suitable tests and confidence intervals for dependent correlations are implemented, for instance, in the R package `cocor` (Diedenhofen & Musch, 2015). Such targeted tests are most informative when the comparisons are specified in advance; in exploratory multiverse designs, the number of possible pairwise comparisons grows quickly, so that summarizing the overall pattern (e.g., by variance components or plots of all conditions) is often more useful than testing every pair. Researchers may be interested in computing logprobs of token predictions to

obtain probability-weighted expected ratings that reflect the uncertainty associated with each token (e.g., Hussain et al., 2024; Martínez et al., 2024).

The estimation uncertainty of some performance measures (such as the correlation of LLM and human ratings) is bounded by the number n_{items} of stimulus items included in the study. The sampling error due to having a finite number of items is relevant with respect to generalizing the results to other items from the target domain (Brennan, 2010). To assess this type of uncertainty, we can average the standard error of an agreement measure across simulation iterations (e.g., the mean SE of the correlation coefficient). Importantly, the averaged standard error depends on the number of items and does not automatically decrease even if more conditions or iterations are used.

A different source of uncertainty in LLM simulations is due to the stochasticity of repeated LLM ratings within the same condition (Knöchel et al., 2025). This uncertainty can be reported in the form of the Monte Carlo standard error (MCSE) and decreases for a larger number of iterations. When studying the quality of LLM-generated ratings, we consider the MCSE to be less relevant than the uncertainty due to studying only a limited number of items, since it depends on a property of the specific procedure (i.e., how many repeated LLM ratings were elicited) and not a property of the content (i.e., whether the rating methodology generalized across the item domain). Siepe et al. (2024) and Morris et al. (2019) provide equations to calculate the MCSE for many performance measures.

Researchers should also indicate how the number of iterations per LLM simulation condition was determined. The choice of a certain number of iterations may be informed by preliminary tests regarding computational feasibility, sample size calculations, or rules of thumb (e.g., conventions). Siepe et al. (2024) provide formulae for sample size calculations in methodological simulation studies that aim at a certain threshold for the maximum MCSE. In certain instances, additional information, such as the variance of a performance measure, is necessary to determine the required number of iterations per LLM simulation condition. However, this information is only attainable after having conducted a pilot study.

LLM output can deviate from the allowed range of responses, resulting in invalid

or missing values. If ratings are to be confined to numerical values in a bounded, predefined range (e.g., Likert ratings between 1 and 7), provide information on how other output (e.g., non-numeric text or values outside of the range) will be treated. One possibility is to handle these values as missing data and report their relative occurrence rate. To reduce such issues, researchers can use LLM software that ensures a certain output structure, for instance, by specifying a specific JSON template (for example with software packages such as ellmer in R; Wickham et al., 2025). If such methods are not used, it may be the case that the rating of interest is still included in the open text output and can potentially be retrieved by further text processing. If this is the case, describe the steps taken to extract relevant information from the open text format. As an alternative to treating invalid responses as missings, it is possible to impute iterations that did not return a valid rating, for instance, by implementing repeated prompting until a valid output rating is obtained (for different strategies of handling missingness in simulations, see Pawel et al., 2026). Furthermore, specify at which proportion of failed iterations a whole condition will be excluded from the analysis.

Finally, one may specify thresholds to reduce post-hoc flexibility in interpreting the results. For example, one could specify what will constitute 'acceptable' and 'unacceptable' levels of performance and what will count as 'relevant differences' in the performance of different LLMs. Still, any verbal labels should be interpreted cautiously, especially if they are based on arbitrary conventions.

Other Information

The last section of the template documents the employed software versions and the computational environment used for the LLM simulation. List the primary packages intended for LLM usage, including their version number. Additionally, include information about the operating system that will be used for running all analyses and for hosting the LLMs. If different computers or servers are utilized for the LLM simulation and the subsequent analysis, report information regarding each of these environments.

After the analysis, the code employed to execute all analyses, as well as the intermediate or complete results of the analyses should be stored in an open online

repository. Supplemental materials may also be provided to ensure a comprehensive documentation of the employed methodology (Argyle et al., 2025). This section can also include information about the most likely obstacles in LLM prompting (e.g., issues with the API or LLM service due to repeated requests) or the simulation design, and plans to overcome them.

Case Study: Evaluating the Quality of LLM Ratings for Valence and Arousal

As a case study, we present a preregistered application of the modified ADEMP template. The simulation study evaluates the quality and robustness of LLM ratings of valence and arousal for multi-word expressions across different models and prompt specifications, building on Martínez et al. (2024), who evaluated GPT-4o ratings for multi-word expressions in five experiments (also covering concreteness). Their Studies 4 and 5 validated GPT-4o ratings of valence and arousal, respectively, against human ratings for two different sets of 96 multi-word expressions, yielding correlations of $r = .95$ (valence) and $r = .92$ (arousal).

We conducted two separate LLM simulation studies—one for valence and one for arousal—since ratings for the two constructs were validated using different item sets. In each simulation, we varied the choice of LLM and the prompt structure by manipulating the required response format and the type of prompt examples provided as context. Assessing both consistency and inter-rater reliability, our analyses focused on the variance decomposition of LLM-generated ratings and the correlation of LLM and human ratings of valence and arousal provided by Martínez et al. (2024).

Before conducting the reported simulation study, we ran preliminary tests to assess the usefulness of the modified ADEMP template and ascertain the functionality of the analysis pipeline. For this purpose, we evaluated the quality of LLM ratings regarding the syntactic complexity of factual statements using the items and human ratings by Schmidt and Heck (2024). The code for this pilot study is available on Zenodo (https://doi.org/10.5281/zenodo.23810473).

Preregistered Evaluation of LLM-Generated Valence and Arousal Ratings

We used the ADEMP template for LLM simulation studies for planning and preregistration (https://osf.io/8e6gf).[3] We report the results for valence in the main text and those for arousal in the supplementary materials (https://doi.org/10.5281/zenodo.23810473).

Aims

The aims of the LLM simulation study are (1) to conduct a variance decomposition of LLM ratings of valence and arousal regarding multi-word expressions into construct variance (due to differences in the valence and arousal of the included items) and method variance (due to specific LLM and prompt variations) and (2) to assess the extent to which LLM ratings correspond to the perceived valence and arousal of multi-word expressions as rated by humans.

Data-Generating Mechanism

Our study uses two different sets of English multi-word expressions (each with $n_{\text{items}} = 96$ items). Martínez et al. (2024) selected the stimuli based on their own GPT-4o ratings to cover the entire range of the respective construct. In Experiments 4 and 5, human raters rated the stimuli regarding valence and arousal, respectively. All analyses were conducted separately for the multi-word expressions of valence and arousal but with identical methodology reported in the Methods section. All LLM calls were independent by starting each LLM call with a fresh context (also see Conde et al., 2026).

We used open-weights LLMs which are run locally through the software Ollama (Ollama, 2026) and are accessed via the R package `ellmer`. To ensure that LLM ratings of valence and arousal only consist of numerical values, we used the `type_object()` and `type_integer()` functions to request output in JSON format with a predetermined response format of integers. Each prompt is composed according to the structure used by Martínez et al. (2024) in prompt 2 (valence) and prompt 3 (arousal) of their paper: (1) An explanation of the task, (2) the range of responses allowed, (3) examples for

[3] The preregistration slightly differs from the current version of the ADEMP template due to improvements made during the review process.

multi-word expressions, (4) the target multi-word expression to be rated, and (5) a reminder for the range of responses allowed.

Estimands and Targets

Our target estimands are the valence and arousal of multi-word expressions (where the scale varies across conditions as detailed in the Methods section in the following). Valence and arousal are the two primary variables of dimensional models of affect (Russell, 1980). Valence refers to the pleasantness of a stimulus, ranging from very negative to very positive, whereas arousal refers to the intensity of the emotional activation elicited by a stimulus, ranging from calm to excited (Warriner et al., 2013).

Methods

We included eight open-weights models by four companies that can efficiently be run locally using Ollama on a GPU with 32GB VRAM without requiring access to an online service. The eight models were gemma3 (with 4b, 12b, and 27b parameters; Google)[4], llama3.1 (8b parameters; Meta), phi3 (3.8b, 14b parameters; Microsoft), phi4 (14b parameters; Microsoft), and gpt-oss (20b parameters; OpenAI). All model parameters were kept at the values that applied by default in our software setup. We deliberately did not tune these settings, because default configurations correspond to how most researchers will use these models in practice, which makes the results informative for typical applications. Although Ollama documents a default temperature of 0.8 for models without a model-specific value (llama3.1:8b, phi3:3.8b, phi3:14b, phi4:14b), accessing Ollama via `ellmer` sets temperature and top-p to 1 for all requests. Hence, all eight models ran with the same temperature and top-p, whereas top-k was 64 for the gemma3 models and 40 for all other models. Appendix B documents all settings, and an additional check in the supplementary materials confirmed empirically that ratings obtained with the default settings corresponded to those obtained with a temperature of 1 rather than 0.8. All models were updated to their most recent version at 28/01/2026.

We adopted the prompt design by Martínez et al. (2024) to obtain either a valence or an arousal rating while manipulating two key components in a fully factorial setting:

[4] The labels indicate that the models have 4, 12, and 27 billion parameters.

(1) The format of the rating task (a 1-9 or a 0-100 scale) and (2) The presence and type of example expressions provided: no example expressions; three examples for low valence or arousal (lower anchor); three examples for high valence or arousal (upper anchor); or three examples for both low and high valence or arousal (both lower and upper anchor). The last condition corresponds to the original prompt design used by Martínez et al. (2024). Ratings of valence and arousal were obtained in two separate and independent steps of the LLM simulation. The system prompt was empty, thus not providing any custom system instructions. Example prompts can be found in the Appendix.

The complete LLM simulation design including all investigated method factors had $8 \times 2 \times 4 = 64$ distinct conditions in total, as summarized by the following list:

- *Model:* eight models from four different companies, run locally using Ollama
- *Format:* response scale, ranging either from 1 to 9 or from 0 to 100, that the LLMs had to adhere to in their response
- *Example:* four types of different examples provided to the LLMs (none, lower anchor, upper anchor, both lower and upper anchor)

Performance Measures

The present investigation used two performance measures: (1) Estimating the variance components in LLM-generated valence and arousal ratings associated with differences in the stimulus materials and differences in methods (secondary performance measure focusing on consistency), and (2) assessing the correlation between human and LLM ratings of valence and arousal (primary performance measure focusing on inter-rater reliability).

Regarding aim (1), we estimated the variance components of the following linear mixed model using the `gtheory` package (Moore, 2016) which uses the `lmer()` function,

```
lmer(llm_rating ~ (1 | stimulus * model * format * example),... )
```

The `stimulus` term accounts for differences in the target construct between the multi-word expressions (i.e., actual variability in valence or arousal), whereas the remaining terms (`model`, `format`, and `example`) account for methods effects.

Following generalizability theory, we first performed a G study, in which the variance components of all random effects (main effects and interactions) were estimated from the fitted model and expressed as percentages of their sum (Brennan, 2001). All variance components were estimated as random effects because each model, example type, and response format is drawn from a potentially infinite universe with the stimuli being the object of measurement. The realization of the chosen model, exact example type, and presentation of the response format are to some extent arbitrary and interchangeable and could differ across studies. Treating any of these facets as fixed would move interactions of the stimuli with these facets to the universe score variance and thereby increase the generalizability and dependability coefficients. The present analysis can therefore be regarded as conservative.

In the D study, the universe score of a stimulus is its expected rating across all admissible conditions of the facets (e.g., across all models and prompt variants that could have been used), and the variance of these scores across stimuli constitutes the universe score variance. We defined it as $\sigma^2(\tau) = \sigma^2(\text{stimulus})$ (i.e., as the true, unknown random-intercept variance of the stimuli), because we aimed to evaluate how well the obtained stimulus LLM ratings generalize across the investigated facets of the design (8 models, 4 example conditions, 2 response formats). The generalizability coefficient

$$E\rho^2 = \frac{\sigma^2(\tau)}{\sigma^2(\tau) + \sigma^2(\delta)} \tag{1}$$

compares the universe score variance to the relative error variance $\sigma^2(\delta)$, which comprises all components involving an interaction with the stimuli and thus affects the relative ordering of stimulus ratings. In contrast, the dependability coefficient

$$\Phi = \frac{\sigma^2(\tau)}{\sigma^2(\tau) + \sigma^2(\Delta)} \tag{2}$$

is based on the absolute error variance $\sigma^2(\Delta)$, which additionally includes facet main effects and their interactions, so that $\Phi \leq E\rho^2$. Our analysis focuses on $E\rho^2$, as our study focuses on whether LLMs can produce stimulus ratings that show the same relative

ordering (but not necessarily the exact same values) as those provided by humans.

The D study variance components were obtained by dividing the respective G study variance components by the product of the number of factor levels per involved facet (excluding the object of measurement – the stimuli). For example, the D study variance component for the interaction of the stimulus s, the model facet m, and the example facet e is given by $\sigma^2(sME) = \frac{\sigma^2(sme)}{n'_m \cdot n'_e}$, where $\sigma^2(sme)$ is the corresponding G study variance component and n'_m and n'_e are the number of models (= 8) and example conditions (= 4), respectively. The capital letters M and E indicate that the estimated variance component is for stimulus mean scores across 8 models and 4 example type conditions. Because $\sigma^2(sME)$ involves an interaction with the stimuli, it is part of $\sigma^2(\delta)$, whereas components without such an interaction (e.g., $\sigma^2(ME)$) enter only the absolute error variance $\sigma^2(\Delta)$.

Regarding aim (2) about the agreement between human and LLM ratings, we computed the Pearson correlation coefficient,

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x}) \cdot (y_i - \bar{y})}{(n-1) \cdot \mathrm{SD}(x) \cdot \mathrm{SD}(y)}. \tag{3}$$

We focused on relative agreement because the case study aimed to identify method factors affecting the relative ordering of stimulus ratings, which is the relevant property when LLM-generated norms are used to order or select stimuli and which corresponds to the validation strategy of Martínez et al. (2024). Moreover, absolute agreement would be partly arbitrary, since ratings from the two response formats (1–9 and 0–100) would first have to be rescaled to the human rating scale. We computed the correlation at two different levels of the analysis:

- (a) At the level of individual iterations in each simulation condition, we computed the correlation between human ratings and LLM ratings across expressions. The inter-rater reliability (IRR) was estimated by the average correlation,

$$\widehat{IRR} = \frac{\sum_{i=1}^{n_{\mathrm{sim}}} \hat{r}_i}{n_{\mathrm{sim}}}. \tag{4}$$

To assess the estimation error due to the finite number of iterations, we computed the Monte Carlo standard error,

$$\text{MCSE}_{\widehat{IRR}} = \frac{\text{S}_{\widehat{IRR}}}{\sqrt{n_{\text{sim}}}}, \tag{5}$$

where $\text{S}_{\widehat{IRR}} = \sqrt{\sum_{i=1}^{n_{\text{sim}}}\{\hat{r}_i - (\sum_{i=1}^{n_{\text{sim}}}\hat{r}_i/n_{\text{sim}})\}^2/(n_{\text{sim}}-1)}$ is the sample standard deviation of the estimate and n_{sim} is the number of simulation iterations (see Siepe et al., 2024).

- (b) At the level of aggregated LLM ratings (averaged across all simulation iterations within each simulation condition), we computed the correlation between human and LLM ratings. Using a bootstrap procedure, n_{est} LLM ratings were first randomly drawn with replacement from the n_{sim} iterations of a simulation condition for each stimulus and then aggregated into an average rating. These average ratings per stimulus were then used to compute the inter-rater reliability between human-generated and LLM-generated ratings. This procedure was repeated n_{sim} times to ensure that the same number of estimates was obtained as when using raw LLM ratings.

To quantify uncertainty due to the limited number of items, we averaged the standard errors of the correlations for the $n_{\text{items}} = 96$ expressions across iterations[5]; to quantify uncertainty due to the limited number of iterations, we computed MCSEs (Siepe et al., 2024). Both types of standard errors and corresponding 95% CIs are reported in plots and as tables in the online materials.

We performed 100 simulation iterations per condition. The number of iterations was determined based on preliminary testing with ratings of syntactic complexity (Schmidt & Heck, 2024), which showed that this number of iterations resulted in narrow MCSE 95% confidence intervals while ensuring computational feasibility. LLM ratings that did not fall within the specified range of admissible rating values (integers either

[5] This was achieved by computing the average lower and upper bound for each correlation across iterations.

between 1 to 9 or between 0 to 100 depending on the condition) were treated as missing values. We report the proportion of such cases and excluded conditions if the proportion of missing values exceeded 20% based on a rule of thumb.

Regarding our primary performance measure, the correlation between human and LLM ratings, we preregistered the following thresholds: an inter-rater reliability of $\widehat{IRR} \geq .70$ was labeled acceptable, $\widehat{IRR} \geq .80$ adequate, and $\widehat{IRR} \geq .90$ (as obtained with GPT-4o by Martínez et al., 2024) excellent. These cutoffs follow common conventions for reliability coefficients, which were proposed mainly in the context of test–retest reliability (Groh, 2025; Matheson, 2019). The specific cutoffs and their verbal labels should therefore not be read as substantive quality categories, but rather serve to constrain post-hoc flexibility in interpreting the results. The continuous estimates and their uncertainty remain the primary basis for interpretation. We considered a certain method as performing better (or worse) than these thresholds on the employed corpus if the MCSE 95% CI was above (below) a threshold. We considered Method X as performing better than Method Y in a certain simulation condition on the employed corpus if the lower bound for the estimated inter-rater reliability of Method X ($\widehat{IRR} - 1.96 \cdot \text{MCSE}$) was greater than the upper bound for the estimated inter-rater reliability of Method Y ($\widehat{IRR} + 1.96 \cdot \text{MCSE}$). Conversely, if the 95% CI based on the averaged standard errors of correlation coefficients between human-generated and LLM-generated ratings was above (below) one of the thresholds, we concluded that a certain method is generally better (worse) than this threshold for comparable stimulus material. As noted above, the MCSE-based CI only reflects the precision of the estimate in a condition, not whether performance generalizes to new stimuli.

Other Information

We used R version 4.5.2 (R Core Team, 2025) for the LLM simulation and the R package `ellmer` 0.4.0 (Wickham et al., 2025) for standardizing the API calls to the locally installed LLMs, which were run via `Ollama` 0.12.9 (https://ollama.com/; Ollama, 2026). We ran the LLM simulation study on a Kubuntu 24.04 machine using an Nvidia GPU (RTX 5090) with CUDA 13.0. For the analyses based on the obtained LLM

simulation results, we used R version 4.6.0 and the following R packages in their most recent versions: `tidyverse` 2.0.0 (Wickham et al., 2019) and `ggplot2` 4.0.3 (Wickham, 2016) packages for data analysis and visualization, `papaja` 0.1.4 for modification of plots according to APA guidelines (Aust & Barth, 2025), and `gtheory` 0.1.2 (Moore, 2016) and `lme4` 2.0-1 (Bates et al., 2015) for conducting generalizability theory analyses. The complete output of `sessionInfo` for all analyses is available at Zenodo (computational_environment.txt; https://doi.org/10.5281/zenodo.23810473). The fully reproducible simulation script and a data set containing all LLM ratings of valence and arousal as well as the correlation between LLM ratings and human ratings with their confidence intervals are available at Zenodo (https://doi.org/10.5281/zenodo.23810473). Furthermore, we generated random seeds for each of the LLM ratings prior to the simulation, thereby ensuring the reproducibility of our analysis.[6]

Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study, and we follow JARS (Appelbaum et al., 2018). All data, analysis code, and research materials are available at https://doi.org/10.5281/zenodo.23810473. This study's design and its initial analysis were preregistered; see https://osf.io/8e6gf/overview. Upon revision of the manuscript and following reviewer comments, we preregistered additional analyses and replaced the fixed-effects variance decomposition with a generalizability theory analysis under https://osf.io/28ja9/. We used Claude Sonnet 4.6 and Opus 5.5 (Anthropic, 2026) and DeepL Write (DeepL, 2026) for grammar checking and linguistic revision and reviewed all proposed changes.

Results for Valence Ratings

Missing Values

Of the 614,400 LLM ratings (64 conditions × 96 expressions × 100 iterations), 38 were missing due to internal errors returned by Ollama during prompt processing (gpt-oss

[6] While testing to see if we would obtain the same LLM ratings for multi-word expressions by repeating a prompt with the same seed, we found that calls to gpt-oss (20b) did not always produce the same output.

[20b]: 37; phi3 [14b]: 1). All other invalid values resulted from violations of the requested response format: gemma3 (4b): 2; gemma3 (12b, 27b): 0; llama3.1 (8b): 263; phi3 (3.8b): 2,216; phi3 (14b): 43; phi4 (14b): 40; gpt-oss (20b): 82. These values were more frequent with the 0–100 scale ($N = 2,451$) than with the 1–9 scale ($N = 195$), suggesting that the finer 100-point format is harder for LLMs to adhere to. By example type, there were 671 (none), 230 (low only), 805 (high only), and 940 (both) invalid values. Overall, missing or invalid values made up only 0.44% of all valence ratings (arousal: 0.06%; see supplementary materials). No condition exceeded 20% missing or invalid values.

Variance Decomposition by Method Factors

To address aim (1), we first rescaled ratings from the 1–9 format to the 0–100 scale to avoid artificial differences between response formats and then fitted the mixed model specified above. Table 1 shows the estimated variance components and their percentages. Since we aimed to investigate consistency—whether the ordering of stimuli was stable across models, response formats, and prompt examples—we were predominantly interested in the interactions involving the stimulus factor, that is, the relative error variance components. For instance, a stimulus × model interaction would indicate that item ratings vary systematically between LLMs.

The stimulus factor accounted for the by far largest proportion of variance (68.85%) in LLM valence ratings. This is desirable, as actual differences in the valence of multi-word expressions perceived by humans (i.e., the construct variance, or universe score variance in generalizability theory, to be measured) should have the largest effect on LLM ratings. Still, Table 1 shows that a considerable portion of the observed variance (almost 20%) was attributable to methodological facets (method variance). The residual variance component (12.84%) was relatively small, indicating limited variability of ratings across iterations within each cell of the design.[7]

In the D study, we estimated the coefficients $E\rho^2$ and Φ for our design with eight model conditions, four example conditions, and two response format conditions. The

[7] Since all ratings were obtained in a fresh context with a new random seed, iterations are exchangeable within cells and therefore allow separating the residual from the four-way interaction.

generalizability coefficient $E\rho^2 = .98$ corresponds to the expected correlation between mean ratings from two independent studies with different facet conditions (Brennan, 2001), indicating excellent reliability. The dependability coefficient was only slightly lower ($\Phi = .96$), indicating that absolute valence ratings for the stimuli (averaged across facets) were reliable. The D study above assumes a single iteration per cell. Beyond the analyses we preregistered, we considered the number of iterations in an additional analysis for another D study, leading to a shrinkage of the residual term, resulting in $E\rho^2 = .99$ and $\Phi = .97$. Given the already high coefficients, accounting for the 100 iterations led to only marginal improvements here, although averaging across iterations generally reduces error variance in LLM simulations (see also the aggregation analysis below). Finally, we conducted another exploratory D study where the number of facets per factor was set to $n = 1$, yielding $E\rho^2 = .75$, indicating the expected correlation between two different LLM-rating conditions across the investigated stimuli (moreover, $\Phi = .69$).

Similar results were obtained for LLM arousal ratings. However, the proportion of variance reflecting universe score variance of stimulus ratings (52.53%) was substantially lower compared to that of valence ratings (68.85%), indicating lower consistency. The effect of this difference was especially apparent when the number of facets was set to $n = 1$ where $E\rho^2 = .75$ for ratings of valence but $E\rho^2 = .58$ for ratings of arousal, indicating that ratings of arousal from a single measurement procedure are not as reliable.

Table 1
Variance components in LLM valence ratings based on a mixed model.

Effect	$\hat{\sigma}^2$	Percent
Stimulus	525.25	68.85
Model	18.39	2.41
Format	3.06	0.40
Example	21.96	2.88
Stimulus × Model	38.82	5.09
Stimulus × Format	0.05	0.01
Stimulus × Example	5.61	0.74
Model × Format	5.82	0.76
Model × Example	10.13	1.33
Format × Example	0.00	0.00
Stimulus × Model × Format	10.72	1.40
Stimulus × Model × Example	14.91	1.95
Stimulus × Format × Example	0.22	0.03
Model × Format × Example	1.00	0.13
Stimulus × Model × Format × Example	9.00	1.18
Residual	97.94	12.84

Note. The universe score variance and the relative error variance components (part of $\sigma^2(\delta)$) are highlighted in bold. The variance component for Format × Example was estimated at the boundary of zero, so lme4 reported a singular fit. Because the model converged, we retained the full model rather than simplifying it as described in the preregistration for convergence problems.

Correlation Between Human and LLM Ratings

To address aim (2), we computed the correlation between human and LLM valence ratings across the 96 stimuli for each of the 64 conditions (Figure 2; the dashed line indicates the correlation of $r = .95$ obtained with GPT-4o by Martínez et al., 2024). As correlations are bounded by the reliability of the variables, we also assessed the reliability of the human benchmark ratings with an analogous generalizability analysis. In the G study, a crossed design with random intercepts for stimuli, raters, and their interaction (the residual) attributed 59.66%, 0.76%, and 39.58% of the variance in human ratings to these components, respectively. In the D study, the generalizability coefficient (cf. Equation 1) was computed as,

$$E\rho^2 = \frac{\sigma^2(\tau)}{\sigma^2(\tau) + \frac{\sigma^2(sr)}{n'_r}}, \tag{6}$$

where $\sigma^2(sr)$ is the stimulus-by-rater (residual) component and n'_r the number of raters. The resulting $E\rho^2 = .96$ (arousal: .95, based on 23 raters) indicates excellent reliability of the mean valence ratings across the 16 raters in Martínez et al. (2024), meaning that the mean item ratings from two independent sets of 16 raters would be expected to correlate at $r = .96$. In this simple crossed design, $E\rho^2$ is equivalent to Cronbach's α computed across persons instead of items (Brennan, 2001).

Figure 2 also illustrates the two types of uncertainty distinguished above. The 95% confidence bounds based on the MCSE (colored error bars) are barely visible, indicating that the correlations were estimated precisely with 100 iterations per condition. In contrast, the averaged 95% confidence intervals (gray shaded areas), which reflect the limited number of 96 stimuli, were considerably wider. However, even accounting for the limited number of items, it is evident that the models differed in their ability to approximate human ratings of multi-word expressions.

Figure 2 shows that the models gemma3 (27b parameters) and gpt-oss (20b parameters) achieved an excellent, albeit slightly lower correlation with human ratings in most conditions compared to the results for GPT-4o reported by Martínez et al. (2024,

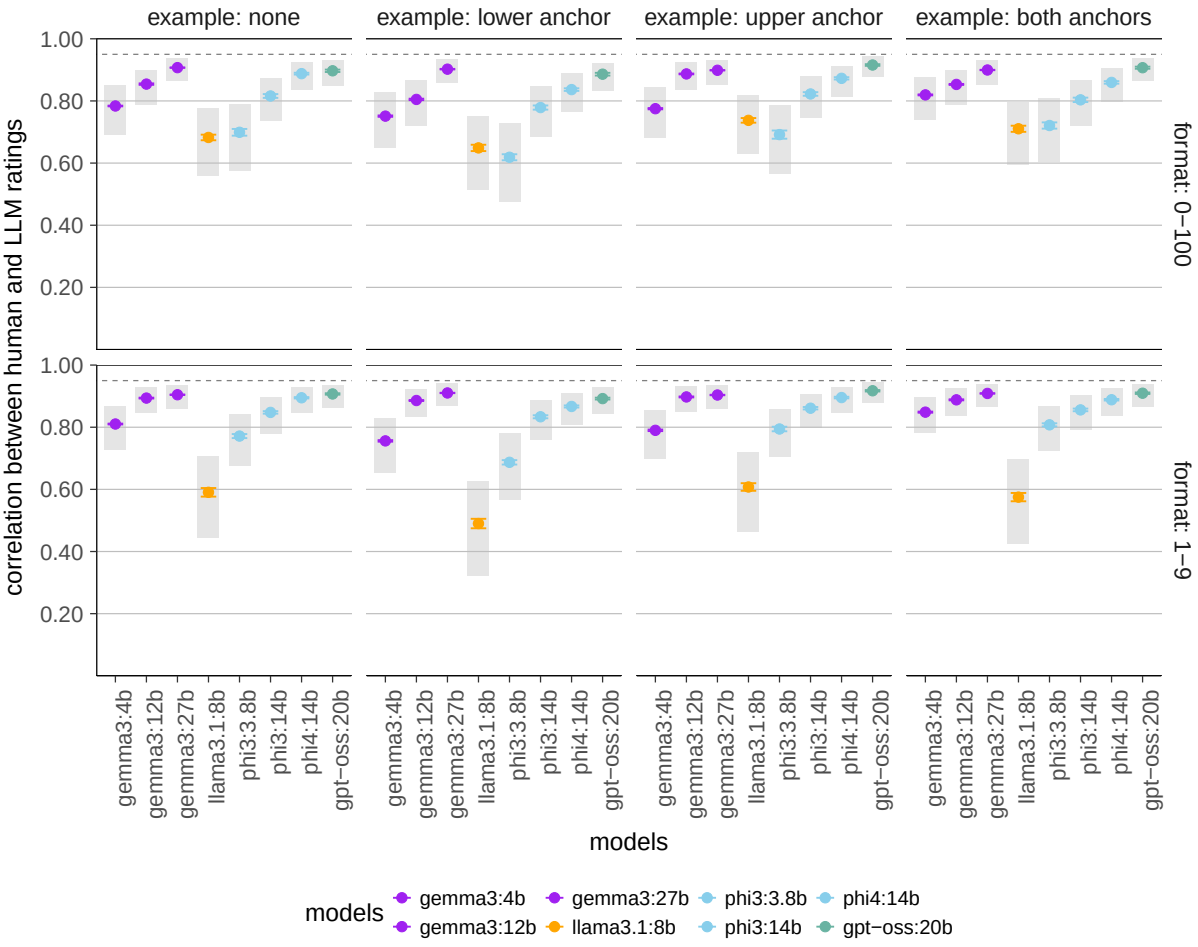
$\widehat{IRR} \geq .90$). Some models such as llama3.1 (8b parameters) and phi3 (3.8b parameters) achieved acceptable correlations with human ratings ($\widehat{IRR} \geq .70$) in some conditions but showed greater dependence on the specific simulation condition. The highest correlation of $r = .92$ between human and LLM ratings was achieved by gpt-oss (20b parameters) using a 1-9 response scale with only high-valence examples in the prompt. Figure 3 provides a scatterplot of human ratings and mean LLM ratings averaged across iterations for each stimulus and simulation condition for the best-performing model. Note that averaging across all iterations before computing the correlation between human and LLM ratings increases the correlation to $r = .95$. Regarding LLM ratings for arousal, we observed similar, but on average lower correlations of up to $r \approx .80$ ("adequate performance"), below the performance of $r = .92$ observed by Martínez et al. (2024) for GPT-4o.

To assess whether the performance ranking of LLM simulation conditions generalizes to new stimuli, we conducted a split-half analysis (exploratory). We randomly split the 96 stimuli into two halves of 48 stimuli each. Within each half, we computed the correlation between human and LLM ratings for each condition and iteration and averaged these correlations across iterations, yielding 64 average correlations per half (one per condition). We then computed the Pearson and Spearman correlations between the two halves across the 64 conditions. Repeating this procedure for 1,000 random splits yielded a mean split-half correlation of $r = .95$ $[.90, .98]$ and a mean rank correlation of $r_s = .90$ $[.83, .95]$ (values within square brackets indicate 95% percentiles). Applying the Spearman–Brown formula to estimate the stability for the full set of 96 stimuli yielded $r = .97$ and $r_s = .95$. Thus, the relative performance of the LLM simulation conditions was highly stable across independent sets of stimuli. However, this stability does not guarantee that the findings generalize to new datasets (see next section).

We further assessed whether the ordering of the 64 conditions in terms of their correlation with human ratings generalizes from valence to arousal (Figure 4). Conditions with higher correlations for valence also tended to show higher correlations for arousal, $r = .68$, $p < .001$ ($r_s = .58$, $p < .001$). As Figure 4 suggests, this consistency was mostly

Figure 2

Correlation between human and LLM ratings of valence for multi-word expressions



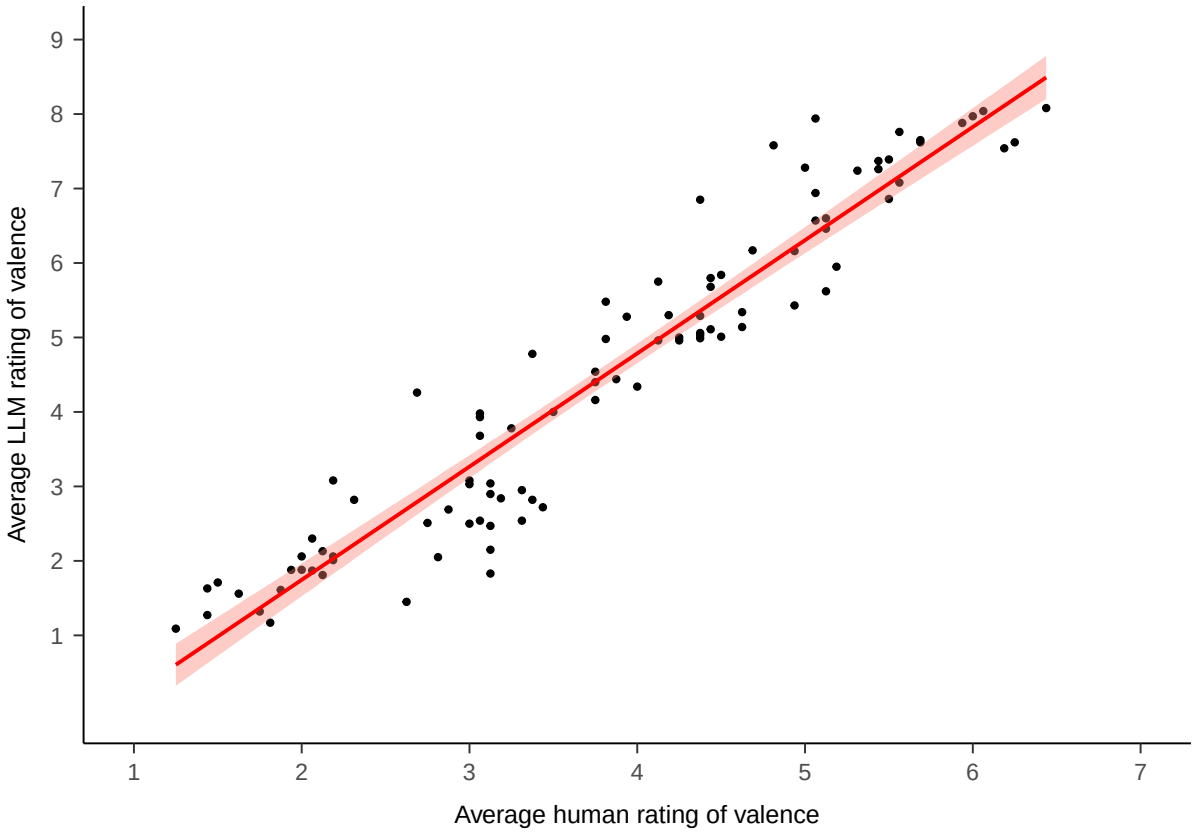
Note. The dashed horizontal line indicates the correlation between human and LLM ratings obtained with GPT-4o by Martínez et al. (2024). Colored error bars (dark gray in print) indicate 95% confidence bounds based on the MCSE. The gray shaded areas indicate the 95% confidence interval for each correlation coefficient, averaged across simulation iterations in each condition.

driven by the models: A model that correlated highly with humans for valence also did so for arousal, whereas differences within models (example type and response format) often did not transfer between constructs.

We also examined whether averaging repeated LLM ratings, which reduces their stochasticity, improves the correlation with human ratings. Using the bootstrap procedure described in the Performance Measures section (level b), we varied the number of averaged ratings $n_{\text{est}} \in \{1, 5, 10, 25, 50, 100\}$, where $n_{\text{est}} = 1$ corresponds to Figure 2. Figure 5 shows the results for the condition implementing the original prompt of Martínez et al. (2024) (1–9 format, examples of low and high valence). Aggregation

Figure 3

Highest correlation between human and LLM ratings of valence for multi-word expressions

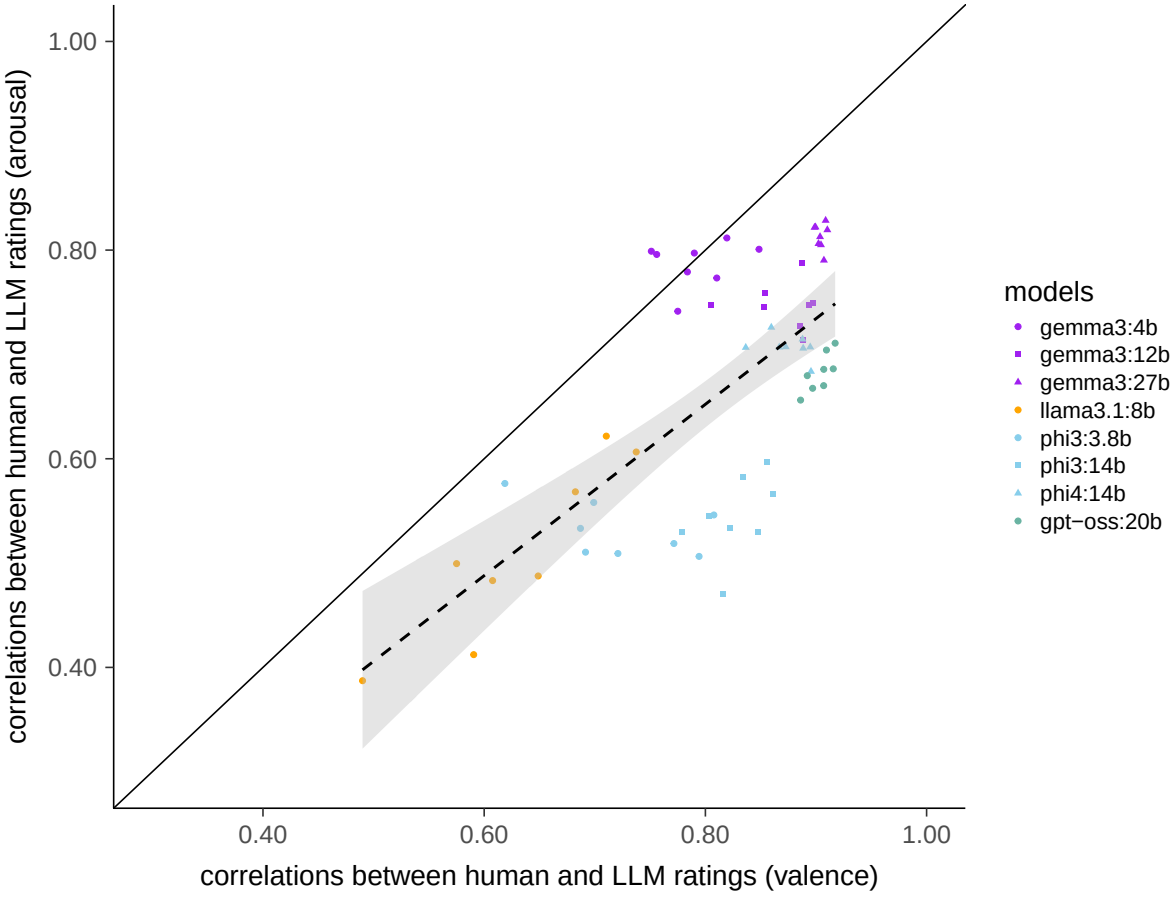


Note. The scatter plot shows the condition with the highest correlation between human and LLM ratings (gpt-oss (20b) with a 1–9 response format and examples at the upper anchor, i.e., high valence). To visualize results, stimulus ratings were aggregated across iterations. The shaded band indicates the 95% confidence interval of the regression line.

substantially improved performance for smaller models such as llama3.1 (8b), whose $\widehat{IRR}$ of approximately .60 to .70 (Figure 2) reached a level comparable to that of larger models when averaging across all 100 iterations. In contrast, gemma3 (especially 27b), phi4 (14b), and gpt-oss (20b), whose unaggregated ratings already correlated highly with human ratings, improved only slightly. Determining the optimal number of iterations is itself a question that generalizability theory can address, as illustrated by the D study above (Brennan, 2001). Across all conditions, the highest correlation after aggregation was obtained for gpt-oss ($r = .95$), matching that of the much larger commercial model GPT-4o used by Martínez et al. (2024).

Figure 4

Comparison of correlations between human and LLM ratings in the simulations for valence and arousal



Note. The black line indicates a perfect match between the LLM simulations for valence and arousal. The dashed regression line illustrates the observed consistency between the LLM simulations, with gray shaded areas indicating the 95% confidence interval.

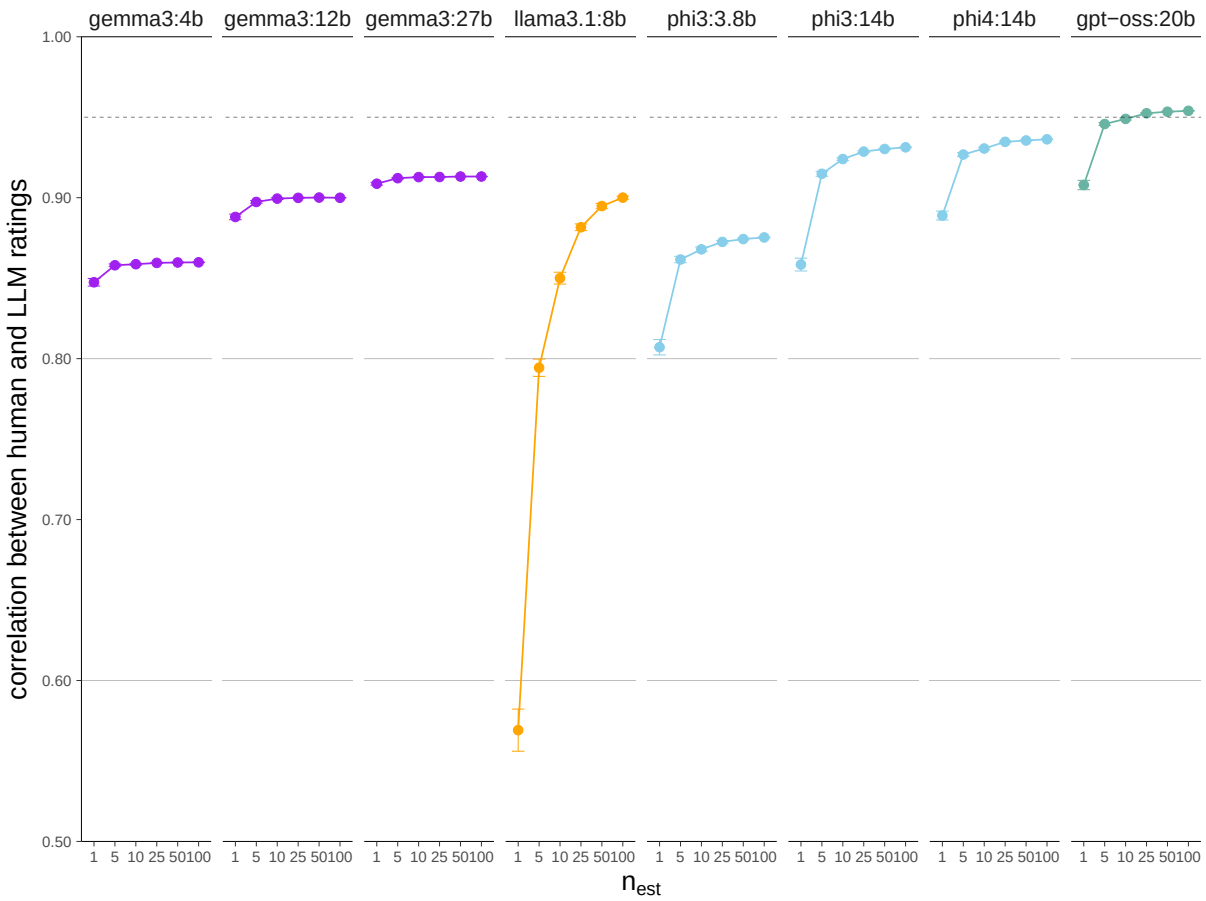
Discussion

The present work proposes a systematic workflow for evaluating the quality of LLM-generated ratings for stimulus materials. We take a novel perspective by connecting LLM-based methodology for natural language processing with traditional methodological simulation studies. Our framework enhances the coherence across disparate fields that have used LLMs to elicit ratings and promotes the adoption of rigorous methodological standards that have been effective in related disciplines. To this end, we adapted the ADEMP framework originally proposed in the field of statistics for simulation studies (Morris et al., 2019) and the corresponding ADEMP-PreReg template (Siepe et al., 2024).

Given the inherent flexibility and numerous, often arbitrary design choices

Figure 5

Correlation between human and LLM ratings of valence for different levels of aggregation



Note. The plot shows the condition with both types of examples and a 1-9 scale. The dashed, horizontal line indicates the correlation between human and LLM ratings obtained with GPT-4o by Martínez et al. (2024). Colored error bars (dark gray in print) indicate 95% confidence bounds based on the MCSE.

involved in generating LLM ratings, a simulation-based multiverse analysis that quantifies the method variance introduced by different models and prompts is often preferable to relying on a single model with a single prompt (Hussain et al., 2024). If only one or a few setups are tested, it remains unclear whether LLM performance generalizes to other materials and procedural variations, which raises the question of whether scientific knowledge has actually been generated (Argyle et al., 2025). Our simulation-based ADEMP framework complements other approaches which emphasize the importance of incrementally developing LLM methods for a specific task of interest (Conde et al., 2026). Besides rigorous method evaluation, the multiverse approach can also be useful for exploring the impact of different methodological factors, such as model

type, hyperparameter settings, or prompt details. Once a strong configuration has been identified for a given task, subsequent evaluations can compare new models or prompting strategies against this best-performing configuration.

Due to the fully crossed design of a multiverse study, its computational costs are much higher than those of a single LLM setup, especially for large stimulus sets such as the norms of concreteness, valence, and arousal generated by Martínez et al. (2024) for 126,397 single words and 63,680 multi-word expressions. These costs can be limited by applying an ADEMP validation study to a random subset of stimuli from the target corpus. Because the subset is a random sample, the findings generalize to the remaining stimuli, with the item-based standard error quantifying the uncertainty of this generalization. The best-performing configuration can then be used to rate the full corpus, for which obtaining human ratings would be infeasible (e.g., Beucler et al., 2026; Martínez et al., 2024). Such validated LLM ratings offer a fast and cheap alternative to human ratings and can be used to sample stimuli representatively from the whole range of a construct for subsequent studies with human participants, as done by Martínez et al. (2024). However, stimulus sets that cover all areas of the response distribution equally, as in the present case study, may increase construct (universe score) variance and thereby lead to an overestimation of the generalizability coefficient.

We illustrated the new ADEMP template by conceptually replicating and extending the validation study by Martínez et al. (2024). In line with their findings, we observed high correlations between human and LLM ratings of valence and arousal. However, with smaller local models, rating quality depended on the chosen model and, to a lesser extent, on the response format and prompt examples, particularly in combination, as quantified by the variance components (Brennan, 2001). A limitation is that subjective human ratings served as the benchmark, which need not be correct and can be distorted by inattentive responding and other response biases (e.g., Stosic et al., 2024). Given the high reliability of the human ratings and the difficulty of measuring valence and arousal otherwise, however, there seems to be no better alternative benchmark. Despite the sensitivity of LLM-generated ratings to methodological choices, even modestly sized,

locally run open-weight models with approximately 8 to 27 billion parameters produced valence ratings that correlated with human ratings comparably to GPT-4o, particularly when ratings were averaged over repeated queries under identical conditions.

Concerning limitations, LLM performance may vary even for related constructs: For arousal, open-weight models reached only adequate performance ($r \approx .80$; see Results), below that of GPT-4o (Martínez et al., 2024). This may partly reflect properties of the construct, since arousal ratings are generally less consistent across norming studies than valence ratings (Muraki & Pexman, 2025; Warriner et al., 2013). Alternatively, the paper and data by Martínez et al. (2024) may have been part of the training data of some models (e.g., gpt-oss and gemma3, published in 2025), which could inflate performance estimates ("leakage"; Feuerriegel et al., 2025; Hullman et al., 2026), possibly more so for valence than for arousal. Ruling out this explanation requires replicating the present investigation with new stimulus materials and human ratings. Finally, other procedural details besides model type, prompt examples, and response scale may affect LLM response quality; while we selected those deemed most relevant, future research is required to study other prompt design choices (Fang et al., 2026).

A fundamental caveat concerns the interpretation of LLMs' predictive performance. While LLM output may show high correlations of up to .95 with human ratings, such high predictive performance does not imply that LLMs have a causal understanding of cognitive processes in humans or that the underlying mechanisms are equivalent (Demszky et al., 2023). Loru et al. (2025) investigated this question and showed that LLMs produced output similar to that of human raters in a news classification task, but humans and LLMs still differed in the processes leading to their conclusions. Furthermore, high correlations between LLM and human ratings do not justify simply replacing human participants with LLMs. Rather, systematic LLM simulations varying procedural details, planned and possibly preregistered with the ADEMP template, allow researchers to investigate where discrepancies between LLM and human ratings may occur and which factors are relevant for minimizing this gap.

When using commercial LLMs as a research tool, the lack of information about

model architecture as well as training data and procedures threatens scientific transparency and reproducibility. Additionally, many large models cannot be run locally and are accessible exclusively through APIs (Feuerriegel et al., 2025; Wulff & Mata, 2025a). In the present study, we employed open-weight models run locally using Ollama (Ollama, 2026), accessed via `ellmer` in R (Wickham et al., 2025), thereby enabling other researchers to easily replicate and extend our analyses. When accessed via programming languages such as Python, open models even allow researchers to access internal model information such as log probabilities of LLM predictions (which reflect the uncertainty associated with each response) or to apply fine-tuning to further increase the correlation with human ratings (Hussain et al., 2024; Martínez et al., 2024). However, open-weights models are not without limitations. For instance, even if the architecture and weights are openly available, the complete set of factors, design choices, and training data relevant to their development may not be known.

Another general issue is the substantial computational and environmental cost of LLMs (Abdurahman et al., 2024; Debelak et al., 2025; Demszky et al., 2023), which can be mitigated by smaller models that run locally on consumer-level hardware. Costs also increase with the context length (i.e., the number of input and output words a model can consider; Hussain et al., 2024), and long inputs may be more difficult for LLMs to process (Huang et al., 2025). In the present investigation, we used concise rating instructions and requested only integer ratings, thus keeping context length minimal.

Finally, LLM output may be biased related to demographic factors such as race or sex. This is especially critical when using LLMs for rating tasks, as LLMs may replicate existing societal biases present in their training data (Abdurahman et al., 2025; Debelak et al., 2025; Guest et al., 2025). By conducting an LLM simulation study using the present framework, researchers can investigate biases. On the one hand, biases not specific to particular items will manifest as changes in the absolute level of all LLM ratings, and thus in a larger main effect of model type on stimulus ratings (e.g., generally higher ratings for one model than another model). On the other hand, model biases specific to certain item properties will manifest as an interaction between model and

stimulus type (e.g., systematically different ratings for items involving men vs. women).[8] For instance, researchers may systematically vary certain terms and content related to specific demographic groups in the stimulus materials (e.g., whether an applicant is framed as male or female) to examine how specific aspects affect LLM ratings.

The proposed template can also be used for more specialized and advanced methods of eliciting LLM ratings. For instance, researchers may forgo LLM prompting altogether and instead extract psychological constructs from text data using embedding spaces provided by certain models. Recent studies have used embedding spaces to analyze semantic relationships between various psychological constructs and item texts (e.g., Herderich et al., 2024; Hommel and Arslan, 2025; Hommel et al., 2026; Wulff and Mata, 2025b). Similarly, fine-tuning can tailor LLMs to specific tasks (Conde et al., 2026; Hussain et al., 2024). Although a multiverse of design choices is not strictly needed for continually improving a fine-tuned model, the ADEMP approach can help evaluate how the choice of base model and hyperparameter settings affects the resulting fine-tuned models.

In conclusion, the present paper contributes to the robust use of language models as a research tool by developing a workflow for evaluating LLM-generated ratings. We adapted the ADEMP template for statistical simulation studies to facilitate the planning and reporting of LLM simulations. The framework draws on generalizability theory to disentangle the contributions of items, model and prompt choices as well as repeated iterations to the consistency of LLM ratings (Brennan, 2001; Cronbach, 1972), while using inter-rater reliability with human ratings as a gold standard for validity (Goodwin, 2001). Overall, our framework provides a psychometric perspective on evaluating LLM output across the multiverse of arbitrary model and prompt choices.

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[8] In generalizability theory, the former rater biases are systematic facet main effects which contribute to absolute error variance, whereas the latter biases are referred to as object-facet interaction variance which is the primary component of relative error variance (Brennan, 2001).

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Appendix A

Prompt Examples

Each LLM prompt consists of five components that are concatenated in each iteration of the simulation:

1. An explanation of the task,
2. the range of responses allowed,
3. examples for multi-word expressions,
4. the target multi-word expression to be rated, and
5. a reminder for the range of responses allowed.

The examples below are based on Prompt 2 provided by Martínez et al. (2024) for ratings of valence. Ratings of arousal were analogous modifications of Prompt 3 (not provided here).

Example prompt with examples of low and high valence (original version of Prompt 2 in Martínez et al.):

Could you please rate how reading the following multi-word expression makes a person feel. Use a scale from 1 to 9, where 1 means very negative, bad and 9 means very positive, good. Examples of words that would get a rating of 1 are pedophile, AIDS, and wreck. Examples of words that would get a rating of 9 are vacation, fantastic, and laugh. The expression is: [insert expression here]. Only answer a number from 1 to 9. Please limit your answer to numbers.

Example prompt with only examples of low valence:

Could you please rate how reading the following multi-word expression makes a person feel. Use a scale from 1 to 9, where 1 means very negative, bad and 9 means very positive, good. Examples of words that would get a rating of 1 are pedophile, AIDS, and wreck. The expression is: [insert expression here]. Only answer a number from 1 to 9. Please limit your answer to numbers.

Examples for multi-word expressions are (taken from Martínez et al., 2024):

- hit rock bottom
- fall in love
- ice cream cone
- drug addict
- starry sky

Appendix B

Documentation of LLM Parameters

This section documents the parameter settings that were in effect when the models described in the paper were run locally using Ollama (version 0.12.9) and accessed via the R package `ellmer` (version 0.4.0). All LLM calls were run successively on a single GPU (RTX 5090, 32 GB VRAM). All statements refer to the source code of these versions; the online documentation of Ollama partly deviates from it (e.g., it lists a default context length of 2048 tokens).

Ollama determines the parameters of each request in three steps, with each step overriding the previous one: (1) Ollama's built-in defaults, (2) model-specific parameters defined in the model's Modelfile, and (3) options passed with the request itself (see function `modelOptions()` in https://github.com/ollama/ollama/blob/v0.12.9/server/routes.go). Ollama's built-in defaults (https://github.com/ollama/ollama/blob/v0.12.9/api/types.go and https://github.com/ollama/ollama/blob/v0.12.9/envconfig/config.go) are:
Ollama defaults:

- num_ctx = 4096 (context length in tokens; set via the environment variable `OLLAMA_CONTEXT_LENGTH`, which we did not change)
- temperature = 0.8
- top-k = 40
- top-p = 0.9
- num_batch = 512 (maximum number of prompt tokens processed per forward pass)

Model defaults (departures from Ollama defaults as defined in the Modelfiles):

- gpt-oss (20b): temperature = 1
- gemma3 (4b, 12b, 27b): temperature = 1, top-k = 64, top-p = 0.95

However, `ellmer` accesses Ollama through its OpenAI-compatible endpoint (`/v1`; see https://github.com/tidyverse/ellmer/blob/v0.4.0/R/provider-ollama.R, line 81). When a request sent to this endpoint does not specify temperature or top-p, Ollama sets both to 1 and passes them on as request options (see function `FromChatRequest()` in https://github.com/ollama/ollama/blob/v0.12.9/openai/openai.go, lines 556-577). Because request options take precedence, these values override both the Ollama defaults and the Modelfile parameters listed above. No other sampling parameters are set by default on this endpoint. In addition, Ollama increases the context length for the gpt-oss model family to at least 8192 tokens unless the total VRAM of the GPU is below 20 GiB (see function `scheduleRunner()` in https://github.com/ollama/ollama/blob/v0.12.9/server/routes.go), which did not apply to our GPU. Consequently, the parameters in effect in our simulation were:

- temperature = 1 and top-p = 1 for all models
- top-k = 64 for gemma3 (4b, 12b, 27b) and top-k = 40 for all other models
- num_ctx = 8192 for gpt-oss (20b) and num_ctx = 4096 for all other models
- all remaining parameters at the Ollama or Modelfile defaults listed above

The supplementary temperature analysis (20 iterations per condition with default settings and temperatures of 0.8 and 1.0) is consistent with these settings; ratings obtained with default settings corresponded to those obtained with a temperature of 1 rather than 0.8.

Note that the documentation of `ellmer` 0.4.0 states that models accessed via Ollama "can only use 2048 input tokens". This note does not reflect the effective context length in our setup (see above). All prompts in our simulation were far shorter than 2048 tokens, so that none of the context limits affected the input. The gpt-oss override further illustrates that effective settings can deviate from documented defaults and should be verified in the source code or empirically. No reasoning effort was specified, so gpt-oss (20b) ran at its default reasoning level.